

ActInf Textbook Group ~ Cohort 5 ~ Session 10 (Chapter 5, Part 1) ~ 11/21/2023

TextbookGroup/ParrPezzuloFriston2022/Cohort_5 | Cohort 5 ~ Session 10 (Chapter 5, Part 1) ~ 11/21/2023 | 57:46

right thank you for joining everyone
it's 11:21 and we're in our first
discussion for cohort 5 chapter
5 before we go to any specific
questions where does anybody want to
begin or just anywhere chapter 5
related or
adjacent that they want to just begin
with
any thoughts on the chapter overall or
like where it sits in the
book or just anything else at the
whole textbook level or chapter 5
level
um yeah I just this is a little specific
but so far I think this might be the
chapter that goes the most
into
precision and that's kind of a I don't
know that it's such a key concept it
seems an active inference that I just
find that really interesting here um I
especially find the discussion of
like like getting more into the
neurobiological aspect of like different
neurotransmitters corresponding to
tuning like different precisions like
dopamine policies and nor adrenaline
Transitions and so on
so
yeah
yeah where whereas often
the material aspects of the brain are
linked to the kind of estimate of the
mean like with rate coding here it's a

connection between neurotransmitters or sometimes called neuromodulators because through Precision modulation they're influencing how different things happen of course recognizing that like even the same molecule because of the receptor being different and the downstream processing and the context being different like it's not even like one neurotransmitter does one thing even in one species brain so of course this is painting with a broad brush but still these are real research programs that have benefited from unifying different experimental results related to a neurotransmitter Under the Umbrella of like a model parameter specifically Precision parameters precision and variance just kind of being like two ways to describe it like how thin or how wide it's like the same thing you're talking about and in chapter five being where any empirical evidence even comes in or any specific non-toy example comes in Oh chapter one overview low and high road chapter four just the math and the details of the generative model so here's the first time that we're even talking about a real specific system which gives a flavor of like the kind of experiments that people have done kind of framings that people take

towards the mammalian nervous system
human nervous system not to say that's
the only system that active applies to
or anything like that but just that
that's where the most empirical work to
date has
happened so to provide just one more
chapter in between chapter 4 describing
the generative model and chapter six how
to build your own chapter five is just
like this kind of empirical Waypoint to
show here's what some research programs
involving chapters one through four look
like okay now chapter six here's how you
would do that in this system or in a
different
system

this first paragraph has a
important idea about the sparsity of the
statistical model and then the sparsity
of the anatomy of the system and in the
brain where those two sparsities kind of
coexist but having a sparser statistical
model means that variables tend to only
influence a few other variables and that
can make things easier to compute or
easier to interpret
statistically now obviously you can
simplify it or sparsify it to a point
where you're missing out in P vectors
then you've sparsified too much but to
the right extent model sparsification
makes it more computable more
interpretable and the second sparsity is
like the anatomical sparsity which is
the fact that although there are a lot
of connections in the brain with the
dendrites and the neurons and all this

and cells really are connected to a lot of other cells in the body they're not connected to all cells and so to the extent that not all things are connected to All Things most directly or in a given way that's the morphological sparsity and the kind of neuromorphic hypothesis is that the sparsity of the substrate has an interplay with the sparsity of the algorithm such that it can be carried out in an energy efficient way

um in terms of the overall structure of the chapter and AI just please raise your hand or write in the chat if you have a um question the overall structure of the chapter pretty much builds up to figure

5.5 figure 5.5 represents these three neural systems or these three motifs one of them is this kind of canonical cortical microcircuitry the sort of thousand brains

uh Hawkins numenta type cortical microcolumn in the prefrontal cortex the second neural system consider and then that's discussed in the in the context of a hierarchical predictive processing architecture the second neural system discussed is the the variance um the dopaminergic

Seesaw in some of the dopaminergic regions of the of the brain in the basal ganglia and so on this balance between on one hand habit driven policy selection where the policy posterior is basically just passing forward the policy prior π and on the other hand the

expected free energy updated policy
posterior where the policy prior gets
like updated with G and then policies
are selected from either the best or
drawn from that expected free energy
updated uh policy space and then the
third neural system considered is this
spinal reflex arc this is like a cut
across a spinal column with sensory
proprioceptive data coming in and a
sort of error differential being
calculated between descending prediction
incoming sensory data and then that
differential basically the sign and the
magnitude of that differential can be
resolved locally through motor action so
it's kind of broadly sketching out a
type of symbolic understanding or
hierarchical
prediction
a motor and policy selection apparatus
that can have different temperature
values and then once the policy has been
selected it is used in a kind of set
point calculation
and then the differential between the
set point and the actual is used to
enact the
policy so the chapter kind of goes
through those three different neural
systems and sites work that goes into
more in
depth but it's an example about how a
subsystem motif is created and then also
how they can be composed and assembled
to start to get to these
more composite synthetic
architectures anyone have like a thought

or question or quote they want to go to
or we can look to the
questions I had a question like in the
uh 5.5 in the figure which you showed uh
what were the what are the elements in
the in the bottom half of
it

like like yeah this this Pi 3
this is all this third yeah this this
areas yeah um do you want to look at
this the basal ganglia section in the
chapter ah okay okay yeah yeah let's
let's look so first part is that
cortical

Motif second part is that neuro the the
sensory motor okay then here's where we
get to the dopamine part the subcortical
structures first they're introducing
some of the empirical gross and fine
histology like which actual parts of the
brain are connected to what and
expressing what subclasses of
dopamine

[Music]

receptors because of the the BIOS
Semitic interpretation of what different
anatomical connections map to like if
it's coming from a proprioceptor we're
just going to like call that appr
proprioceptive input they use some of
those empiric
structurings to associate
with different um aspects of the
model um it just they're like discussing
even though the the the imag is later
down yeah yeah then so they describe it
then they talk a little bit about
Parkinson's

and about time

scales in terms of what

um that image is

showing and those images are like uh the

the they're like anatomically repres

representative of the like basil gangli

the stum something like

that okay

yeah um I'm I'm going to go to a figure

where the the beta and gamma are shown

clearly okay so here's step-by-step

tutorial from Ryan Smith this is model

stream discuss the model stream

one

um okay so um this step by step builds

up to the model at the bottom right from

first static perception so prior hidden

State mapping between states and

observations observations so thermometer

in the room truth true underlying hidden

temperature prior on temperature mapping

between thermometer and the temperature

in the

room then the B Matrix is introduced

this is a different pedagogical path

than the textbook takes which why it's

good complimentary resource then from

here with have the same L then B is the

transition Matrix so that's how the

temperature and the room actually

changes time to time as modeled um could

stay the same could

change this is still a passive inference

problem and this is like the music

listening example from chapter 7 in the

textbook then the upstairs control

component is introduced π is policy

policy intervenes in the transition

Matrix it basically selects a transition
Matrix to use it's like as many policies
as there
are they're like index cards and then
one of them is selected out and that's
the Matrix um that's used for the
transition under that policy policy
selected by this expected free energy
calculation on the policy prior and part
of that the pragmatic value component is
driven by the
preference and then finally on the
bottom right it's kind of like a a
slightly Fuller version now there's no
fullest version because you could
continue to add uncertainties and all
these other things to all the parameters
so it's not like the stopping point but
here it's more explicit that the policy
is um driven not just by expected free
energy based evaluation but but also
it's driven by the prior on policy here
it's just being shown more clearly and
then here's that like variance on policy
and it it's just saying that that gamma
is like one over beta so just saying
whether you think of it as like being
high Precision or low
variance or high variance slash low
Precision those are the same it's like
saying how light is this how heavy is
this it's like you're getting the same
value so what that variance or Precision
on policy is doing is it's taking the G
updated policy posterior there what
functionally what it's doing and it's
sharpening or
flattening that resulting distribution

of policy um
likelihoods so under the high
temperature setting low Precision High
variance policy sampling setting then um
differences amongst policies are blurred
so that's one dimension that that policy
selections controlled that is whether
or
in the low temperature setting whether
the policy that's slightly more likely
is always selected so that's the
temperature component and then there's
within
um um let's just say lower temperature
decision making there's this question of
how much does
g sharpen P_i ranging from none at all
policy prior just getting passed through
to high
level sharpening it a lot so a policy
with even a little bit more or a little
bit better expected free energy little
bit more expected um pragmatic or
epistemic value becomes like greatly
differentiated from other
policies sanj's textbook has some really
good visualizations of like how these
features influence each
other
Nikos uh what was that textbook and uh
also have a question about one of the
diagrams um sanjie Nam josi's active
inference textbook like kind of in
preparation if anybody wants to see it
then then then email and let us know um
but but he has some
new new ways to lay out some of the
spaces of those like like that kind of

space of policy like high and low temperature and then like all all those things but those are like some of the knobs that exist on that diagram that you have before I think it was of the pdps with increasing complexity uhuh yeah so just to confirm so are these each letter is representing an expectation Matrix or some those are all matrices and that would be I guess I'm trying to understand if all of these are internal to the system

that's doing the modeling good

question the environment is not shown here these are all describing the agent itself and then the difference between the square and the circle they're all variables in the computer program so but um however as shown the the squares are kind of like the ones that tend to be kind of stated and fixed and then the circles are the are the ones that are getting like that are being that where inference is happening like State inference is happening on them but that's not an in principle thing it's actually just sort of an implicit graphical style they might describe it in the caption But ultimately they're all just variables okay got it thanks

in the lowest level I think those were sigmas little tiny sigmas isn't when it's observation doesn't it touch on the environment or it's still we consider it

still it was on the previous picture you were showing eer

like those the lowest the the yeah

observations yes the O's yes like this

is the thermometer reading at time point

one two and three so it's it's on the

robot but it does relate to the

environment so you're right it is coming

from the environment this is kind of the

where um these are the sense States so

to speak but they're on board they're

not this is not modeling the

environment and can I ask something um

ber deis was mentioning common filters

would that would that be that area that

you were just showing the observations

like kind of be modeled as a common

filter or am I off track

here yeah it's a great so here's from

7.1 it's the only place I see where um

common filter is mentioned in this Koda

so like here's the bottom left there

here's the seven figure 7.1 so it's same

exact thing so common filter um like we

could look it up and there's a ton of

things to go into but a common filter is

basically solving the inference problem

how much Decay should the signal have

how much time averaging and how should

the signal Decay through time so that

we're not like over under smoothing so

it's very related to to generalized

filter

it's just like how much should we smooth

this signal to do optimal reconstruction

on

so you could interpret you you could

interpret this as doing a

colan estimation through time probably
getting in a sequence of observations
and needing to do an underlying State
estimate through them that's smoother
we're getting noisy estimates from the
thermometer

and then we're going to like draw a line
of best fit basically through
them or you could interpret these that
would be over a very F that would be
like a fast time scale or maybe we're
talking about the hidden state every
um every you know minute and now so it's
slower and now we're just thinking okay
the a matrix

summarizes a kind of smooth
so that we're we're not you know it's
probably different ways to do
it let's see if it's in the
textbook chapter

part of smoothing part of optimal
smoothing is estimating how variable it
is so that's another connection with the
Precision

estimate if it's highly variable you
need to do a lot of
smoothing if it's low variability you
don't need to smooth that
much

and then there's some technical points
about whether like the fluctuations of
the stochastic process are this way or
that

here's here's one one case where the
Precision on policy
matters

um that they bring up on page

92 so here there's two policies that are

possible like staying seated and being standing just to simplify and if we're seated then we're getting sensory input that we're seated we're getting confirmatory evidence we have high Precision on the everything is kind of lined up the ball is kind of at the bottom of a bowl but how do we consider the adjacent possible sensory data that we would get from being standing such that that movement to standing can actually be realized the reward learning approach there's there's different approaches so not not not to oversimplify but would be something like you have to switch something in the model so that standing is more rewarding and then learn or decide upon a policy by which that more rewarding policy is selected so kind of just pushes up a question to how do you um come to assign that standing state with higher reward um here the the which is discussed in the context of like standing and seating um it's also discussed in the context of icting and precision on the sensory input and and also especially in literature Precision on like more um cognitive parameters is associated with with a wide range but the the movement and the catic are these are

kind of like gross
versions and it's kind of this common
pattern where um in advance of a
movement being made there's an
attenuation of precision on the sensory
input and then that enables it to become
plausible that
um the sensor data are this alternative
way like the variability of the of the
estimate on the incoming sense data
includes both how it actually is and how
it could be and then from that kind of
like foot in the door
possibility that is leveraged to
actually select that possibility like
the the the select action shift is made
while the foot's in the door and then as
the Precision is clamped back down it's
like oh yeah it's totally likely and
possible and confirmatory now that we're
standing so that's this kind of sensory
attenuation phenomena so that's kind of
like attentional focusing and then
attenuation is like the opposite of
attention or just describing like the
the diffusion or the suppression of
attention which is to say increasing the
variance but that's an example about how
whether it's like a Precision on sensory
or a precision on policy like there's a
lot of different
knobs to play
with even without the kind of reward or
habit or learning
knobs but just in the Precision knobs a
lot of phenomena can
arise but these are kind of the cores to
statistical elements we return to

estimates like expectations Center

Tendencies distributions

variances and

differentials like

here

inks that's a silly comment but I was

thinking if someone can make a board

game where we block out different the

different knobs and we only play with

some knobs that would be cool yeah if we

can just play with it and see what

happens

yeah I I I I hope we do have physical

and um digital games I I have several

times experienced when people doing

somewhat like a

guided thought Journey with the goal to

increase or reduce Precision on

different things call are there digital

games that do

this um

like none that I specifically know of

there's Pablo at all's digital game

related to ACM that they presented at

the the Symposium but but that wasn't

explicitly like a psycho technology

modulating

Precision but let's find out and add to

this

list wi with all the strengths and

weaknesses in caveats of kind of

entering at the phenomenological level

like just our experiences on any given

moment are are

not equivalent to the technical

definition of some variance estimator

but again it gives us a really good

experience and sometimes it really does

line up

and like it it does point to to to

really important

phenomena and it helps helps yeah yeah

Nikas uh sorry to interrupt I was just

um G to ask about figure

5.1 thought that was pretty dense um I

was wondering if you could skim over it

and uh in particular um just highlight

again the distinction between the

subscript V and

X in the middle

column

okay

anyone want to give a thought on this

first I'm just going to bring in here so

we can see it little

easily

okay so there's the six cortical layers

just just so that people see the

hystological part of it there these

these six kind of semi morphologically

defined layers in the

cortex this is not a six layer

hierarchical predictive processing

architecture actually all six of them

Implement like one hierarchical node

which is then Associated laterally into

something like higher order basian

um nested models just to be clear we're

talking about like a six layer

integrated unit not a six layer

hierarchical model but they are

layers um okay the

left side summarizes these literature

cited and um the cell types are

described like with the

sssp and this is just describing the

actual cell populations and their
connections the middle message passing
that underwrites hierarchical predictive
coding so there's two
um kinds of nodes here there's Mew for
um let's let's double check but I
believe Tilda is representing as it
usually does through
time or generalized coordinates I'm not
sure
um V and
X I just want to find where X is defined
or used in the
text I think the last time I saw it was
back in chapter 4 I don't know if he re
restated it
yeah where X and V were
used yeah
I I would go with the MU and x and V
definitions from for unless it looks
like
they're
changed
um X in this situ so here here in in 415
is kind of a classical SPM
style um two equation
setup the second line is our
neuroimaging device we're getting sensor
data y real scientist sensor data y it's
a function of underlying neural activity
and V this kind of control unobserved um
control State plus sensor noise and then
change in the neural activity through
time is defined by um it stay at the
current moment and then and then this
causal Force V plus a variability
associated with with
it what would an example of a causal

force be like the attentional
state as far as I understand this from
SPM so this would be like the neural
activity at a given time so here's the
Bold signal or the um EEG signal X would
be the the um the actual neuronal
activity which is what what um
synchronously connects to a given sensor
measurement and then this would be like
the like attention or no
attention unfoldings of that
system okay I I

I believe this is accurate now return
here but we should confirm and we should
annotate the figure if someone puts in
the the the research and feels confident
and then please annotate the figure so
that we don't need to do ad
hoc interpretation because I'd like to
have the ontology
based just simple description of this to
actually go to rather than need to forge
for

it's it's an interesting progression
that they go
through because on the
right is where we see the kind of O and
S PDP

type um variables whereas in the middle
is the message passing more chapter four
variables and would be good to
understand if if where are onetoone
assertions um or concordances being
claimed and what would the
interpretation of that
be we can look to the papers to see
more but on the
incoming our sensory

areas our information flow is coming
in and then kind of policy relevant
auction information is coming
out modeled by the a matrix and the the
hidden States coming
in and the
at a lower level so this is the kind of
hierarchical logic lower level State
estimates coming
in and then at the ith level the state
estimates being done here
influencing policy
selection but a lot more remains to be
connected about how these are how this
is formally linked to to you know for
and
elsewhere uh do you mind if I take a
stab
at let if it is in the ballpark so um X
could be maybe for example like glucose
metabolism
or um you know how cells are regulating
the glucose metabolism and V would be
some
underlying maybe thought process
um governing where
attention
I guess thought process that is being
influenced by you know some deeper
Dynamics going
on in a nervous
system would that be kind of an
accurate so those are those processes
are both
happening in the the uh cortex but
they're representing two
physical processes that are influencing
each

perhaps hard to evaluate just in in the moment but it would be cool to flush it out
okay cool thanks yeah thank you no it's good we we we should have good accounts of all the figures

Pros

yeah I just uh was trying to understand the diagram and uh they were also mentioning that uh uh information is flowing from uh lower cortical areas and it's this directionality it's the like this information is going to the layer four and then uh from three it's going to higher layers so we have that flow of information which can also help us in understanding that yeah just wanted to point that out that's it

yeah here's where a three a three- layer basian model is being shown kind of the lower the middle and the higher level and each of these levels they're sending up um

observations that are being juxtaposed with descending predictions within a and for each

X and for V there is a mean and there's a mean and and an a differential estimators it could be the mean in the variance

and that's not even incompatible with it being um the um neural activity and the regime of

attention under the model that that the variance is what is

controllable high and low variance condition which is exactly this kind of

neuromodulator as internal
policy par saying well this is the knob
that you can
control you can release or or not
release and you know modify the
sensitivity of the post synaptic neuron
and the downstream receptor and all that
but you can't just be like well this
neural population is firing
at intensity 4 let's just make its
intensity nine because there isn't a way
for the Post synaptic neuron necessarily
to send back and and to know what to ask
for or to make that retrograde ask on
the right time
scale
so different synaptic
mechanisms modify the
Precision so that different neural
messages can be tuned in and tuned
out brains a complex system even as as
mentioned in um 56.2 today
um like even the nematode and the fully
mapped nervous systems of the fruit fly
and the
honeybee have a lot of
sophistication not that it's like a
ladder of of complexity or anything like
that but
just these are big
big evocative simplified
models however they are really developed
out and used in
Precision neural settings and where
there are important critiques and and um
extensions those are literally research
and development
advances but the chapter tries to

represent like a little bit of a
higher symbolic
generalized a little bit of a more basic
or Elementary cognitive reflex arc and
then the kind of like in between with
the dopamineergic policy selection um
Andrew um yeah just for
like I guess like clarification and
maybe there's not enough information
here to be able to fully answer it but
like on 98 with um back to the different
neurotransmitters or I guess neuromodulators um so like for example with
serotonin there at the bottom of the
list and preferences or interoceptive
likelihood being like the Precision
that's that it's being modulated in that
case with like a way we could say that
is that serotonin is um
changing
the it's
so it's modulating the Precision meaning
it's like
adjusting
basically
how how confident one can be in
um in the interoceptive likelihood such
as someone is currently experiencing I
don't know like a increased heart rate
and there's like a signal coming in like
informing them of that or their
breathing is faster or or you know
they're hungry right now or something um
a higher
Precision like if this is really
simplifying everything but like say
serotonin were the key lever here that
says okay the serotonin is released it

modulates the Precision of a hunger
signal such that you
now pay more attention I suppose to that
hunger signal or it appears stronger or
rather the the likelihood or Precision
of like I'm feeling this
sensation uh I'm certain I'm like very
confident that it means I'm hungry like
is that kind of how that is playing out
in some f way that wasn't too
much that's a good question also they
mentioned preferences or interceptive
likelihood but of course they're not the
same thing but let's let's think about
what that would be yeah
to have a precise preference
distribution would be like I really want
the Die to come up as one and I really
don't like two through six imprecise
would be like kind of flat across
preference distributions so what would
it be like to have the flat or sharp
interceptive likelihood distribution
well if it were totally
flat I'm getting ambiguous signals from
interception I'm not
I'm not getting um useful or
epistemically meaningful information
about whether my heart rate is high or
low high
Precision I I'm I'm super
sure I'm super sure that I know what my
inter receptive state
is could be a false Precision on the
extreme case
too but this is dealing with the
parameter right I don't even think
using at least

not in this

chapter what one one reason why the

precisions

matter okay so here's figure 7.10 so

here's the kind of figure

4.3 that we usually see

PDP and now the a b c d and e the

variables that you specify like in

chapter six to have the agent those all

have these lowercase distributions on

top of them they're they they're priors

on the other

distribution

um and so that prior distribution So

like um let's just say it's a coin flip

setting we could say um I I have my

prior is like one zero it started on

heads period or your prior could be 5050

0. 5.5 like um I I I'm equally confident

that it started on one or the other

state that's without the lowercase D so

now we add in say we're going to draw

our prior on D across trials we're going

to draw that from 0. 5.5 plus or minus

0.1 so then one trial you you you start

with a prior 6.4 then 0 4.6 and 0. 5.5

and so on so you're drawing from a prior

about D and then that could be narrow or

it could be

tight and then just like in any other

basian statistical setting like when a

distribution is wider it can be moved D

more the distributions narrow even

incredible evidence is going to be

basically systematically downweighted so

severely that the the distribution only

is going to move a little

bit

um even though the evidence May
repeatedly be
signaling that it's off in the extreme
case if you have like the Delta
distribution just like a spike you say
no the average um height of the children
in the classroom is 5 ft plus or minus
zero then even getting other values
doesn't update your
distribution but these are like variance
parameters you can think of on the other
ones because these are basically again
to return to those simple statistical
motifs we can basically be talking about
the central tendency of a distribution
like the the median or the mean the
average expectation leloy and
approximation all these things so
central tendency
mean variance or
variability those are kind of the two
parameters as part of like the
generalized gaussian scheme and then a
simple Motif is like a differential
between two
means but a big part of the
Simplicity of the hierarchical
predictive processing architecture is
all it does is calculate means es and
differentials and that there's no
parameter that's loaded up with like
Mega
significance and taking on like too much
more than just describing a a mean
variance or differential for a
given model
feature
nios ju so just to clarify were you

saying that the variance in for example
a normal distribution is is equivalent
to the

Precision in this
sense

um there like just just it's in the
spirit of I know that there might be
like a little bit of technical detail if
it's a co-variance or something like
that but yes the the standard deviation
a larger standard deviation is larger
variance but I know there's like
standard deviation and variance
but yes at a broad scale just talking
about how fuzzy versus sharp a
is high Precision gaussian low variance
low standard
deviation low Precision High variance
big standard
deviation and the flatter the curves
basically the more leeway you have in
picking whatever the policy whatever the
curve represents policy is or beliefs or
whatever yeah limit case uniform
distribution like everything's equally
likely then like whatever data point you
get is like as good is the best one that
you have yeah the other limit case the
Delta distribution whatever data point
you get you go with what you already
thought can you go back to the diagram
that you had up just a second
ago just had a quick question about that
was it I think it was it was the mdp
with the um
ABCD oh oh okay
okay um it so and this is just
to kind of really try to understand this

model a little bit better is there any particular point in this model and I know it's a little bit different for the one with the squiggly lines from um chapter 4 but is there any particular point point in here where variational or expected free energy is being calculated that we could point to

yes 100% of those calculations of expected free energy are happening at the G Okay g is a functional it's a node that's taking in habit and preference and policy prior and it's outputting the policy posterior the whole upstairs is the decision-making

apparatus the whole downstairs is just hidden state

observations mapping prior so expected free energy never enters into the downstairs

component but that downstairs sense making could be described by variational free

energy okay got it

so that's really helpful and is there some implied intuition here that the there's a feedback loop going on I guess that just happens during message passing iterations feedback loop

what uh you know there's a lot of diagrams especially earlier in the book that are going from perception to action perception

action yeah great great Point great Point here the policy selection it just kind of selects an internal index card and then it's like but then that's

actually Chosen and emitted to the niche
and then that results in the next
observation coming in differently or not
but yes that is not shown the like the a
Markov blanket or the relationship
between the agent and the niece are not
shown that's helpful yeah and and again
son jev's textbook I I I hope some he'll
share some of these images or people can
can look into it because he has some
representations that that that dispel a
lot of the inertia of how things were
visualized not all of it but
but helped a lot um inks and then any
other like last thoughts people
have I'm turning us back I think to
Precision modulation I
was thinking of last time Andrew was
saying these are just constraints and I
was trying to imagine what constraints
would look like with Precision
modulation but I was imagining this game
I don't know if you have it in America
like warm warmer warmer hot cool colder
like as you're approaching an object
stent right like like I imagine the do
is doing that like giving you a warm
warm warmer towards how right you are
about something you're
predicting that's that's awesome like
that's very much like the kind of um
veilance in the sophisticated active
inference that modeled veilance like
psychological veilance positive or
negative they Associated that exactly
with Precision estimates if the
Precision if things are as variable as
you expect them to be you have neutral

veilance if things are more precise than
you expect them to be you have positive
veilance if things are more surprising
than you expect them to you have
negative veilance and anxiety and then
it's like the signal that that is given
by by veilance which is like the top
level you know which way is the wind
blowing of the whole body is like is it
getting warmer am I getting warmer or
not is my free energy on the right
track on the lukewarm expected or or
whatever someone wants to calibrate
their path too and that's why and that's
why they were positing multiple axes of
positive and negative veilance right
it's not positive to negative it just
like it's Precision evaluating different
predictions you have so I imagine
sometimes you predicting something
you're not happy about but the Precision
is kind of making you oh I'm right so
there's like a a bad positive veilance
there if oh yeah um I'm
I'm I'm as expected that I'm the kind of
person who sees scary
movies that surprise me but I'm not
surprised in that I'm surprised that I
feel this way I mean those are the
higher
order um Yarns to unwrap these are just
the
elemental kind of
Legos thank you everybody beautiful
thank you so
much this is
fun times as we kind of as as the as the
niche continues to evolve around us and

we come close to our last discussion in
the end of the year so thank you fellow
see you next time bye thank you thank
you